

TRACK 4: AUTOPILOT AGENT

BusinessFlow AI

Project Plan & Technical Specification

Email-to-Action Business Workflow Automation Agent

Hackathon	Global AI Hackathon Series with Qwen Cloud
Track	Track 4 — Autopilot Agent
Project Name	BusinessFlow AI
Core Technology	Qwen Cloud LLM APIs + Alibaba Cloud ECS
Submission Deadline	July 9, 2026
Developer Background	Full-Stack / Backend Engineer
Estimated Build Time	~20 hours (part-time)

Comprehensive Technical Document · July 2026

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1 · Executive Summary

BusinessFlow AI is an intelligent workflow automation agent that monitors a business email inbox, classifies incoming requests, routes them to the appropriate handler, drafts responses or downstream actions, and pauses for human approval before executing consequential decisions. Built on the Qwen Cloud LLM platform and deployed on Alibaba Cloud, it demonstrates exactly what Track 4 (Autopilot Agent) rewards: ambiguous input handling, external tool invocation, human-in-the-loop design, and production-grade engineering.

Problem

Business teams waste hours manually triaging emails, creating tickets, drafting quotes, and scheduling — all repetitive, low-value work.

Solution

An AI agent that reads, classifies, and acts on business emails end-to-end, pausing only when human judgment is genuinely required.

Differentiator

Production-first design with robust error handling, confidence scoring, and transparent reasoning — not a toy demo.

Tech Foundation

Qwen Cloud chat models + MCP tool integrations + Alibaba Cloud ECS + FastAPI backend + React approval UI.

2 · Problem Statement & Market Context

Modern business teams receive a constant stream of emails, messages, and requests that require repetitive, structured responses: customer inquiries, quote requests, support tickets, vendor communications, and scheduling. A typical operations or sales team member spends 2–3 hours per day on email triage alone — time better spent on higher-value work.

Pain Points This Agent Solves

Manual Email Triage	Humans must read, understand, and classify every incoming email before taking any action. This is slow, error-prone, and doesn't scale.
Inconsistent Responses	Without automation, response quality varies by individual — different tone, different information, different speed.

Context Loss	When someone new picks up a thread, they have to re-read the entire history. Agents retain full context automatically.
Missed SLAs	Time-sensitive requests (quote deadlines, urgent support) get buried in overflowing inboxes.
Siloed Systems	Creating a CRM record, scheduling a meeting, and sending a reply are three separate manual actions — an agent does all three atomically.

3 · Solution Overview — BusinessFlow AI

How It Works — End-to-End Flow

1

Inbox Monitor

The agent polls the email inbox at a configurable interval (e.g., every 5 minutes). New emails are fetched and queued for processing.

2

Intent Classification

The Qwen LLM classifies each email into one of several intent categories: Customer Inquiry, Quote Request, Support Ticket, Vendor Communication, Scheduling Request, or Unknown.

3

Confidence Scoring

Each classification is assigned a confidence score (0–1). Low-confidence classifications (< 0.75) are flagged for human review automatically, regardless of intent type.

4

Context Enrichment

The agent queries the CRM to retrieve existing customer records, previous interactions, and open tickets for the sender. This context is injected into the next step.

5

Action Planning

Based on intent + context, the agent produces an action plan: which tools to call, in what order, and what the draft response should say.

6

Human-in-the-Loop Gate

Before executing any consequential action (sending email, creating record, scheduling meeting), the action plan is sent to the Approval Dashboard for human review.

7

Execution

Once approved (or auto-approved for low-risk actions), the agent executes each tool call: `draft_email()`, `create_ticket()`, `schedule_meeting()`, `update_crm()`.

8

Audit Logging

Every action, decision, confidence score, and tool call is logged to a structured audit trail for transparency and debugging.

4 · System Architecture

Architecture Philosophy: Modular, stateless where possible, and observable. Each component has a single responsibility. All external calls are wrapped in retry logic. The system degrades gracefully — if the CRM is unreachable, the agent continues without enrichment.

Component Map

PRESENTATION LAYER	
React Approval Dashboard	Single-page web app. Shows pending agent action plans. Humans click Approve / Reject / Edit. Connected to backend via WebSocket for real-time updates.
Status Monitor UI	Read-only view of processed emails, decisions made, and audit log. Useful for demonstrating agent behavior in the video demo.
APPLICATION LAYER (FastAPI, Alibaba Cloud ECS)	
Agent Orchestrator	Core loop: fetch emails → classify → enrich → plan → gate → execute. Stateless between runs; all state stored in Redis/DB.
Intent Classifier	Calls Qwen Chat API with a structured classification prompt. Returns intent label + confidence score + reasoning.
Context Enricher	Queries mock CRM API to retrieve customer history and open tickets for the sender's email address.
Action Planner	Given intent + context, generates a structured action plan as JSON: [{tool, params, requires_approval}, ...]
Approval Gate	Evaluates each planned action against approval rules. High-risk or low-confidence actions → human queue. Others → auto-execute.
Tool Executor	Executes approved action plans by calling registered tools in sequence with error handling and rollback on failure.
Audit Logger	Writes structured logs (JSON) for every decision, tool call, and outcome. Stored in Alibaba Cloud OSS.
TOOL LAYER (MCP-compatible)	
draft_email()	Calls Qwen to generate a professional email reply given the intent, customer context, and user preferences.
create_ticket()	Writes a structured support ticket to the mock helpdesk system with priority, category, and summary.
update_crm()	Updates the customer record with the latest interaction, status, and extracted preferences.

<code>schedule_meeting()</code>	Creates a calendar event using mock calendar API with auto-detection of time preferences from email text.
<code>send_email()</code>	ALWAYS requires human approval. Sends the drafted reply via SMTP. The most consequential tool.
<code>escalate()</code>	Flags an email as requiring human attention directly. Used when confidence is below threshold.

INFRASTRUCTURE LAYER (Alibaba Cloud)	
Alibaba Cloud ECS	Ubuntu Linux VM hosting the FastAPI backend. Chosen for simplicity of deployment and full control.
Alibaba Cloud OSS	Object Storage for audit logs, email attachments, and generated artifacts.
Redis (ECS-hosted)	Job queue for email processing tasks and session state for the approval workflow.
Qwen Cloud API	External LLM API calls for classification, action planning, and email drafting. All calls metered against \$40 credit.

5 · Core Components & Technical Design

5.1 Intent Classification — Prompt Design

The classifier uses a structured few-shot prompt to ensure consistent JSON output. The prompt is versioned (v1, v2, etc.) and stored in a config file so improvements can be tracked across runs.

```
SYSTEM: You are a business email classifier. Analyze the email and return JSON only. Return: {
  "intent": , "confidence": <0.0-1.0>, "reasoning": , "urgency": } Intent labels: CUSTOMER_INQUIRY |
QUOTE_REQUEST | SUPPORT_TICKET | VENDOR_COMM | SCHEDULING | COMPLAINT | UNKNOWN Rules: - confidence <
0.75 → set intent to NEEDS_REVIEW - Extract sender tone (frustrated, neutral, positive) into reasoning
- urgency HIGH if email mentions deadlines within 24 hours USER: [EMAIL SUBJECT]: {subject} [EMAIL
BODY]: {body} [SENDER HISTORY]: {crm_summary}
```

5.2 Action Planning — JSON Schema

The action planner receives the classified intent and CRM context, then produces a structured plan that the Tool Executor can follow step by step. Each step includes a risk level to drive the approval gate.

```
{ "plan_id": "uuid-v4", "email_id": "msg_123", "intent": "QUOTE_REQUEST", "confidence": 0.92, "steps":
[ { "step": 1, "tool": "draft_email", "params": { "tone": "professional", "include_pricing": true },
"risk_level": "medium", "requires_approval": true, "reasoning": "Quote reply with pricing info – needs
human review" }, { "step": 2, "tool": "update_crm", "params": { "status": "quote_pending", "tag":
"hot_lead" }, "risk_level": "low", "requires_approval": false } ] }
```

5.3 Approval Gate — Decision Logic

Condition	Gate Decision	Rationale
Tool = send_email	ALWAYS requires approval	Sending email is irreversible and customer-facing
Confidence < 0.75	Full plan requires approval	Low confidence = uncertain classification
risk_level = high	Step requires approval	High-risk actions need human judgment
risk_level = medium	Step requires approval	Medium risk flagged for review by default
risk_level = low AND confidence >= 0.75	Auto-execute	CRM updates, logging — safe to automate

5.4 Error Handling Strategy

Qwen API timeout

Retry 3× with exponential backoff (1s, 2s, 4s). After 3 failures, escalate to human queue.

Low-confidence classification	Auto-route to NEEDS_REVIEW, notify human, continue processing next email.
Tool execution failure	Roll back completed steps if possible. Log full error. Notify admin via email.
CRM unreachable	Continue without enrichment. Log warning. Reduce confidence by 0.1 as penalty for missing context.
Malformed Qwen response	Attempt JSON repair using regex. If still malformed, fall back to rule-based classification.
Human approval timeout (> 1 hr)	Escalate to secondary approver or auto-reject the plan and flag for manual handling.

6 · Qwen Cloud Integration Strategy

Qwen Cloud is the backbone of all intelligence in BusinessFlow AI. The integration is designed to maximize credit efficiency while demonstrating sophisticated use of the platform's capabilities.

API Usage Plan & Credit Budget

API Call Type	Est. Tokens/Call	Calls/Demo	Purpose
Intent Classification	~500 tokens	20–50	Classify each incoming email
Action Planning	~800 tokens	15–40	Generate structured action plan
Email Drafting	~600 tokens	10–30	Generate professional reply text
Ticket Summarization	~400 tokens	10–20	Create concise ticket summaries
Ambiguity Resolution	~300 tokens	5–10	Clarifying questions for unclear emails

Credit Management: With \$40 in Qwen Cloud credits, the demo budget allocates \$20 for development and testing, \$10 for the demo run, and \$10 as contingency buffer. Token usage is logged on every call to prevent overrun. Model defaults to qwen-turbo for classification (cheaper) and qwen-plus for drafting (higher quality).

MCP Tool Integration

BusinessFlow AI implements the Model Context Protocol (MCP) pattern for tool registration, enabling clean separation between the LLM reasoning layer and the execution layer. Each tool is defined with a JSON schema that Qwen uses for structured function calling.

Tool Name	Schema (simplified)	Description
draft_email	<pre>{ "to": str, "subject": str, "tone": enum, "key_points": list[str] }</pre>	Generate a professional email reply
create_ticket	<pre>{ "priority": enum, "category": str, "summary": str, "customer_id": str }</pre>	Create support ticket in helpdesk
update_crm	<pre>{ "customer_id": str, "status": str, "notes": str, "tags": list[str] }</pre>	Update customer record
schedule_meeting	<pre>{ "attendee_email": str, "duration_min": int, "preferred_times": list }</pre>	Schedule a calendar meeting
send_email	<pre>{ "draft_id": str, "approved_by": str, "timestamp": str }</pre>	Send an approved email draft

escalate	<pre>{ "reason": str, "urgency": enum, "assigned_to": str }</pre>	Flag for immediate human attention
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7 · Alibaba Cloud Deployment Plan

Proof of Deployment Requirement: The submission must include a link to a code file in the GitHub repo demonstrating use of Alibaba Cloud services and APIs. This section documents all Alibaba Cloud touchpoints to satisfy this requirement.

Alibaba Cloud Services Used

Elastic Compute Service (ECS)	Primary backend hosting. Ubuntu 22.04, 2 vCPU, 4 GB RAM. Hosts FastAPI app, Redis, and Nginx reverse proxy. Region: Singapore (closest to Qwen Cloud).
Object Storage Service (OSS)	Stores audit logs (JSON), email attachments, and generated email drafts. Bucket: businessflow-audit-logs. Public-read disabled.
Alibaba Cloud SDK (Python)	Used for programmatic OSS uploads from the FastAPI backend. Credential managed via environment variables (no hardcoded keys).
RAM (Resource Access Management)	Create a limited-permission RAM user for the app. Only OSS PutObject and GetObject permissions. Principle of least privilege.

Deployment Steps

1. Provision ECS instance (Singapore region) via Alibaba Cloud console
2. SSH in and install: Python 3.11, pip, Redis, Nginx, git
3. Clone repo: git clone [https://github.com/\[username\]/businessflow-ai](https://github.com/[username]/businessflow-ai)
4. Set environment variables: QWEN_API_KEY, OSS_ACCESS_KEY, OSS_SECRET_KEY, OSS_BUCKET
5. Install Python dependencies: pip install -r requirements.txt
6. Start FastAPI: uvicorn main:app --host 0.0.0.0 --port 8000
7. Configure Nginx to reverse-proxy port 80 → 8000 and serve React build
8. Create OSS bucket and configure CORS for frontend uploads
9. Verify: curl [http://\[ECS-IP\]/health](http://[ECS-IP]/health) returns {"status": "ok", "qwen": "connected"}
10. Capture deployment URL and Alibaba Cloud console screenshot for submission proof

8 · User Experience & Human-in-the-Loop Design

The human-in-the-loop (HITL) design is a core feature, not an afterthought. Track 4 explicitly rewards agents that incorporate HITL checkpoints at critical decision points. BusinessFlow AI implements a deliberate, user-friendly approval workflow that demonstrates this capability clearly.

Approval Dashboard — Feature Spec

Pending Actions Queue	Real-time list of action plans awaiting approval. Each card shows: email summary, detected intent, confidence score, and proposed actions. Updated via WebSocket — no page refresh needed.
Action Plan Cards	Each card displays the full plan: which tools will be called, in what order, with what parameters. A reasoning field shows WHY the agent made each decision.
Edit Before Approve	Humans can edit the draft email, change tool parameters, or remove steps before approving. Edited plans are re-logged with the human's modifications noted.
One-Click Approve / Reject	Clear, prominent buttons. Rejected plans ask for a rejection reason (stored in audit log for model improvement).
Auto-Escalation Timer	If an approval sits pending for > 1 hour, a notification is sent to the secondary approver. After 2 hours, the plan is auto-rejected and flagged.
Audit Log View	A separate read-only tab showing all decisions made (automated and human-approved), with timestamps, tool outcomes, and confidence scores.

9 · 20-Hour Build Schedule

Hours 1–2 Setup

- Create Qwen Cloud account, get API keys, verify \$40 credits
- Scaffold FastAPI project structure: /agent, /tools, /api, /models
- Set up GitHub repo with MIT license, initial README
- Create requirements.txt: fastapi, uvicorn, openai (Qwen-compatible), redis, boto3-alibaba

Hours 3–4 Alibaba Cloud Deployment

- Provision ECS instance on Alibaba Cloud (Singapore region)
 - SSH in, install dependencies, deploy hello-world FastAPI endpoint
 - Create OSS bucket for audit logs
- ★ MILESTONE: Proof of Alibaba Cloud deployment ready (get this early!)

Hours 5–7 Core Agent Logic

- Build email poller (mock IMAP or sample email JSON files for demo)
- Build Intent Classifier module with Qwen API integration
- Build Action Planner with structured JSON output
- Write unit tests for classifier (3–5 test emails with expected outputs)

Hours 8–10 Tool Layer

- Implement all 6 MCP tools (mock implementations with realistic outputs)
- Build Tool Executor with sequential execution and error handling
- Build Approval Gate with configurable rules
- Implement Redis job queue for async processing

Hours 11–13 Approval Dashboard (React)

- Scaffold React app with Tailwind CSS
- Build Pending Actions Queue component with WebSocket connection
- Build Action Plan Card component with Approve/Reject/Edit
- Build Audit Log view (read-only table)

Hours 14–15 Integration & Testing

- End-to-end integration test with 10 sample emails
- Fix edge cases: malformed Qwen responses, API timeouts
- Test all 6 tools in sequence on a realistic quote-request scenario
- Verify Alibaba Cloud OSS logging is working

Hours 16–17 Documentation & Architecture Diagram

- Write comprehensive README: setup, usage, architecture explanation
- Create architecture diagram (Excalidraw or draw.io → PNG → embed in README)
- Write code comments for all non-trivial functions
- Add .env.example file (never commit real keys)

Hours 18–19 Demo Video Production

- Plan 2.5-minute video script (see Section 14)
- Record screen capture: show 3 complete email-to-action flows
- Narrate: intent classification → action plan → approval → execution
- Upload to YouTube (unlisted or public), copy URL

Hours 19–20 Submission & Blog Post

- Fill in all Devpost submission fields (text description, repo URL, video URL, architecture diagram)
- Write blog post (dev.to or Medium): 600–800 words, include architecture diagram and code snippets
- Submit on Devpost before July 9, 2026 2:00 PM PT
- Double-check all links work and repo is public with visible license

10 · Judging Criteria Alignment

Every design decision in BusinessFlow AI maps back to the judging criteria. The table below shows how each scoring dimension is addressed.

Criterion (Weight)	How BusinessFlow AI Addresses It	Score Target
Innovation & AI Creativity (30%)	MCP-compatible tool layer; confidence scoring with dynamic approval gate; multi-model strategy (qwen-turbo for classify, qwen-plus for drafting); JSON schema validation on all Qwen outputs; ambiguity resolution sub-agent	High
Technical Depth & Engineering (30%)	Modular FastAPI architecture; Redis job queue; WebSocket real-time updates; full error handling with retries; audit logging to OSS; environment-based config; unit tests; rollback on tool failure	Very High
Problem Value & Impact (25%)	Email triage automation is a universal, high-ROI business problem. The agent is described as open-source with clear API interfaces, enabling community extension. Scenarios shown (quotes, tickets, scheduling) are immediately recognizable to any judge.	High
Presentation & Documentation (15%)	3 complete flows demonstrated in video; architecture diagram in README; Approval Dashboard shows agent reasoning visually; comprehensive documentation with setup instructions; real-time status monitor makes agent behavior transparent	High

11 · Tech Stack & Dependencies

Component	Technology	Version / Notes
LLM Provider	Qwen Cloud (Alibaba Cloud)	qwen-turbo + qwen-plus models
Backend Framework	FastAPI (Python)	v0.110+, async throughout
Task Queue	Redis + RQ (Redis Queue)	Self-hosted on ECS
Frontend	React 18 + Tailwind CSS	Vite build, served by Nginx
Real-time	WebSockets (FastAPI native)	Approval dashboard updates
Cloud Compute	Alibaba Cloud ECS	ecs.c6.large, Ubuntu 22.04
Object Storage	Alibaba Cloud OSS	audit logs, attachments
Reverse Proxy	Nginx	Port 80 → FastAPI :8000
Testing	pytest + httpx	Unit tests for classifier and planner
Code Quality	ruff (linter) + black (formatter)	Enforced via pre-commit hooks
Version Control	Git + GitHub	Public repo, MIT license
Diagram Tool	Excalidraw	Architecture diagram export as PNG

12 · Submission Checklist

- Project built using Qwen Cloud models (Track 4: Autopilot Agent)
- Public GitHub repository with MIT license visible in About section
- Text description of features and functionality on Devpost
- Proof of Alibaba Cloud deployment: link to code file using Alibaba Cloud APIs
- Architecture diagram (PNG) in README and on submission form
- Demo video ≤ 3 minutes uploaded to YouTube with public URL
- Track 4 selected on Devpost submission form
- All README setup instructions tested by a fresh clone of the repo
- No API keys or secrets committed to the repository
- Blog post published and URL included in submission (bonus prize)
- Submission saved and finalized before July 9, 2026 2:00 PM Pacific

13 · Risk Register & Mitigations

Risk	Likelihood	Impact	Mitigation
Qwen API credits run out during demo	Medium	High	Pre-record all demo flows. Keep \$10 buffer credit. Use mock responses as fallback.
ECS instance goes down before judging	Low	High	Take ECS snapshot after successful deploy. Include fallback localhost instructions in README.
Qwen returns malformed JSON	Medium	Medium	JSON repair function + regex fallback. Catch all parse errors explicitly.
Time runs out before video production	Low	High	Video scripted in advance (Section 14). Record early on Day 4 while there's still buffer time.
Feature scope creep over 20 hours	High	Medium	Strict time-boxing per schedule. Cut features before cutting quality on core flow.
Blog post not written in time	Low	Low	This is bonus-only. Draft outline during builds. Write at Hour 19 if time permits.

14 · Demo Script & Video Plan

The demo video must be under 3 minutes. Judges are not required to watch beyond 3 minutes, so every second counts. The script below is timed to 2:30 to leave a comfortable margin.

0:00 – 0:15 Hook & Problem Statement	Open on the Approval Dashboard. Say: 'Every business team drowns in email. BusinessFlow AI reads, classifies, and acts on your inbox — automatically, with you staying in control.'
0:15 – 0:50 Flow 1: Quote Request (Happy Path)	Load a sample quote request email. Show the agent classify it (intent: QUOTE_REQUEST, confidence: 0.94). Show the action plan appear in real time: draft email → update CRM. Human clicks Approve. Show the draft email and CRM update complete. Narrate the confidence score and reasoning.
0:50 – 1:20 Flow 2: Ambiguous Email (Low Confidence)	Load an unclear email. Show the agent classify it (intent: NEEDS_REVIEW, confidence: 0.61). Show it auto-route to the human queue with a flag. Human edits the plan, selects the correct intent, approves a modified action. Narrate: 'When the agent isn't sure, it asks. It never guesses on your behalf.'
1:20 – 1:50 Flow 3: Support Ticket with Urgency	Load a frustrated customer email marked HIGH urgency. Show the agent detect urgency, create a ticket, and flag for immediate human review (escalate tool). Show the escalation notification on the dashboard. Narrate: 'High-urgency signals are caught automatically. Nothing slips through.'
1:50 – 2:15 Architecture Walkthrough	Switch to architecture diagram. Briefly narrate: Qwen Cloud for intelligence, FastAPI on Alibaba Cloud ECS for orchestration, OSS for audit logs, React dashboard for human approval. Show the GitHub repo and deployment URL.
2:15 – 2:30 Closing & Impact	Return to the dashboard showing 3 completed flows in the audit log. Say: 'BusinessFlow AI turns hours of email triage into minutes — with full auditability and human control at every critical step. Built on Qwen Cloud.'

Recording Tips: Use OBS Studio (free) at 1920x1080. Record audio separately with a decent mic if possible. Edit in DaVinci Resolve (free) or iMovie. Upload as MP4 to YouTube (unlisted is fine). Test the link in an incognito window before submitting.

Remember: You have until July 9, 2026 at 2:00 PM Pacific. Get the Alibaba Cloud deployment done in the first 4 hours — it's the hardest requirement to satisfy at the last minute. Everything else can be iterated. Good luck!